BMC Research Notes

RESEARCH NOTE Open Access

Immunomodulatory effects of high versus low intensity diode laser as adjunct to non-surgical periodontal therapy in periodontitis

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Abstract

Objective Periodontitis is characterized by a dysregulated inflammatory response mediated by proinflammatory cytokines. This study aimed to compare the immunomodulatory effects of high-intensity (DLT-H) versus low-intensity (DLT-L) diode laser therapy as adjuncts to conventional scaling and root planning (SRP) in patients with periodontitis.

Results Twenty-seven participants contributing 81 non-adjacent periodontal sites were randomly assigned into three treatment groups: SRP alone, SRP + DLT-H, and SRP + DLT-L. Gingival crevicular fluid (GCF) was collected at baseline, 1 month, and 3 months post-treatment to quantify levels of IL-1β, IL-6, IL-8, and TNF-α. All treatment modalities led to significant reductions in pro-inflammatory cytokine levels over time ($p < 0.01$). DLT-H produced the most pronounced and sustained decreases, with early and long-term suppression significantly greater than both SRP alone and DLT-L. DLT-L yielded moderate yet consistent reductions across all cytokines, whereas SRP alone demonstrated limited and less durable immunomodulatory effects. Pairwise comparisons highlighted the superior efficacy of DLT-H in both early (baseline to 1 month) and late (1 to 3 months) intervals, particularly for IL-6 and TNF-α. These findings indicate that high-intensity diode laser therapy provides superior and modulatory benefits and may enhance the outcomes of nonsurgical periodontal therapy.

Keywords Periodontitis, Diode laser therapy, High-intensity laser, Low-intensity laser, Cytokines, Immunomodulation, Gingival crevicular fluid, IL-1β, IL-6, IL-8, TNF-α

Background

Oral biofilms consist of highly organized microbial communities embedded within an extracellular polymeric substance (EPS) matrix that facilitates bacterial adhesion, interspecies communication, and resistance to host defenses. Biofilm formation involves initial adhesion to the acquired pellicle, followed by microbial co-aggregation and maturation into a complex microbial ecosystem. In periodontal disease, dysbiotic biofilms are characterized by increased abundance of key periodontal pathogens, including Porphyromonas gingivalis, Tannerella forsythia, and Treponema denticola, whose persistence

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promotes host inflammatory responses and periodontal tissue destruction [1].

One of the main causes of tooth loss globally is periodontitis, a chronic inflammatory disease of microbial origin that gradually erodes the gingiva, periodontal ligament, and alveolar bone [2]. A dysregulated host immune response is the main factor driving the disease's progression, even though anaerobic Gram-negative biofilm started it [3].

Cytokines are central mediators linking innate and adaptive immunity in periodontal inflammation [4]. Among them, IL-1β is one of the most extensively investigated pro-inflammatory cytokines in periodontal research due to its pivotal role in inflammatory cell recruitment, connective tissue destruction, and alveolar bone resorption. Furthermore, genetic polymorphisms within the IL-1 gene cluster have been associated with differences in host susceptibility and inflammatory responsiveness to periodontal disease [5].

IL-1β, secreted by macrophages and dendritic cells, promotes osteoclast activity and tissue degradation. Elevated IL-1β, TNF-α, IL-6, and IL-8 in gingival crevicular fluid (GCF), saliva, and serum strongly correlate with disease severity [6–8]. IL-6 induces acute-phase responses and osteoclastogenesis, neutrophil recruitment, and amplification of inflammatory pathways [7, 9, 10]. Cytokine profiling is therefore a valuable tool for evaluating disease activity and therapeutic response [11–14].

The gold standard for nonsurgical treatment is still SRP, which successfully lowers the microbial burden [15]. However, SRP alone is insufficient to eradicate subgingival pathogens, particularly those within soft tissues, and does not adequately modulate host inflammation. Persistently elevated cytokines post-SRP may impair healing and increase recurrence risk [8]. Consequently, adjunctive approaches aimed at both microbial control and host modulation are under investigation.

Several laser systems have been investigated as adjuncts to SRP, including diode, Nd: YAG, and Er: YAG lasers. Er: YAG lasers are highly absorbed by water and hydroxyapatite, making them effective for root surface debridement and calculus removal, whereas Nd: YAG lasers exhibit deeper tissue penetration and strong absorption by pigmented tissues, enhancing bacterial reduction within periodontal pockets. Diode lasers are characterized by their affinity for hemoglobin and pigmented bacteria, providing effective soft-tissue decontamination, hemostasis, and ease of clinical application. In addition, diode laser systems are generally more compact and cost-effective than other laser modalities, supporting their widespread use as adjunctive periodontal treatment options [16–18]. Diode laser therapy has emerged as a promising adjunct due to its antimicrobial and biostimulatory effects. DLT-H produces pechotothermal effects enabling sulcular debridement, epithelial ablation, and bacterial reduction. When combined with SRP, it has been associated with improved clinical outcomes and reductions in GCF cytokines such as IL-1β and IL-6 [16–18]. DLT-L, or photobiomodulation, operates at sub-ablative levels to stimulate repair, collagen formation, angiogenesis, and immunomodulation without thermal damage [19]. Clinical studies demonstrate that DLT-L decreases IL-1β, IL-6, IL-8, and TNF-α, supporting enhanced periodontal healing [13, 20, 21].

Despite these benefits, heterogeneity in laser protocols, wavelengths, and treatment parameters has produced inconsistent results. Consequently, no clear consensus has yet been established regarding the optimal laser parameters for adjunctive periodontal therapy. Moreover, Direct comparisons of DLT-H and DLT-L remain limited, underscoring the need for well-designed trials to establish standardized, evidence-based guidelines for their use in nonsurgical periodontal therapy [22, 23].

This study aimed to compare the immunomodulatory effects of high- and low-intensity diode laser therapy as adjuncts to SRP in periodontitis patients, with the null hypothesis that no significant differences exist between the two modalities in clinical outcomes or cytokine modulation.

Methods

Study design

The study was conducted in Yemen. Ethical approval was granted by the UST Research Ethics Committee, Sana'a, Yemen. All procedures complied with the Declaration of Helsinki and CONSORT 2010. Written informed consent was obtained from all participants. The study evaluated the effect of SRP alone or combined with DLT-H or DLT-L on GCF cytokine levels (IL-1β, IL-6, IL-8, TNF-α).

Sample size calculation

Twenty-seven patients (mean age 33.6 years) contributed 81 periodontal sites, randomly allocated to SRP, SRP + DLT-H, or SRP + DLT-L. Inclusion criteria were age 25–58 years, generalized periodontitis (2018 classification), probing pocket depth PPD from 5 to 9 mm in ≥ 30% of sites, and ≥ 20 functional teeth. Exclusion criteria included smoking, qat chewing, pregnancy, systemic diseases, medications affecting immunity, recent antibiotics or periodontal therapy, and the presence of prostheses or orthodontic appliances [24]. Baseline data were collected, and all samples were obtained by a calibrated examiner.

Sample size was calculated using IL-1β concentration as the primary outcome [25]. Assuming an SD of 38.3 pg/mL, effect size of 33.51 pg/mL, $\alpha = 0.05$, and 80% power, the minimum required was 21 sites/group. To account for attrition, 27 patients with three non-adjacent sites each (total 81) were recruited.

Three non-adjacent sites per patient were labeled A–C and randomized (computer-generated sequence) to one of three groups: (1) SRP, (2) SRP + DLT-H, or (3) SRP + DLT-L. Allocation was concealed in opaque envelopes opened after baseline sampling. Both participants and assessors were blinded to treatment assignment.

Treatment protocol

To standardize baseline oral hygiene conditions and minimize potential confounding factors, every participant had complete mouth supragingival scaling and polishing, as well as standardized oral hygiene instructions. Subgingival SRP was performed with Gracey curettes under local anesthesia and supragingival debridement was performed using ultrasonic scalers. Adjunctive diode laser therapy was applied using a 980 nm diode laser (LASOTRONIX, Poland): DLT-H: 2.0 W, continuous, contact mode for microbial decontamination and epithelial debridement. DLT-L: 0.5 W, continuous, non-contact mode for photobiomodulation and immunomodulation. Protective eyewear was used throughout. No systemic antibiotics or antiseptics were prescribed.

Data collection procedures

GCF was collected from designated sites at baseline (T0), 1 month (T1), and 3 months (T2). Sterile paper points (size 30) were inserted into the crevice for 30 s, pooled, and stored at − 70 °C until analysis.

Laboratory analysis

Samples were eluted in phosphate-buffered saline, centrifuged, and analyzed using ELISA kits (BT-Laboratory, Jiaxing, Zhejiang, China) for IL-1β, IL-6, IL-8, and TNF-α. Assays were performed in duplicate with standard curves generated for each cytokine. Results were expressed as pg/mL of GCF. Standards and samples were assayed in duplicate according to the manufacturer's instructions, and absorbance was measured at 450 nm with a microplate reader. Concentrations were interpolated from standard curves and expressed as pg/mL for comparison across groups and time points [18].

Table 1 Gender distribution, treatment type, and treatment mouth side of the periodontitis patients under study

Parameter	Category	*N*	%
Gender	Female	19	70.4
	Male	8	29.6
Treatment Type	SRP	27	33.3
	DLT-H	27	33.3
	DLT-L	27	33.3
Oral cavity side	Upper right	18	22.2
	Upper left	16	19.8
	Lower right	22	27.1
	Lower left	25	30.9

DLT-L Low-intensity diode laser, *SRP* scaling and root planning, *DLT-H* high-intensity diode laser

Statistical analysis

Data were recorded in Excel, verified, coded, and analyzed using SPSS v25.0. Continuous variables were expressed as mean ± SD, and categorical variables as frequency and percentage. Normality was assessed before analysis. Within-group changes across T0, T1, and T2 were evaluated using repeated-measures ANOVA or the Friedman test. If significant, post hoc pairwise comparisons were performed as appropriate. Between-group differences (SRT, DLT-H, DLT-L) were analyzed using one-way ANOVA or the Kruskal-Wallis test. If significant, post hoc comparisons were used as appropriate. The statistical significance level was set at $p < 0.05$.

Results

Twenty-seven patients (19 females, 8 males; mean age 33.6 years, range 18–58) contributed 81 non-adjacent periodontal sites. These were evenly assigned to three groups: SRP alone, SRP + DLT-H, and SRP + DLT-L. Distribution across quadrants was balanced, and baseline cytokine levels (IL-1β, IL-6, IL-8, TNF-α) showed no significant intergroup differences ($p > 0.05$) (Table 1).

Across all groups, cytokine concentrations progressively declined from baseline to 3 months. Although significant reductions were observed following SRP alone, both adjunctive laser protocols resulted in greater improvements, particularly DLT-H, which demonstrated the most consistent and sustained reduction across all evaluated cytokines. DLT-L also produced significant reductions, though less consistently than DLT-H as shown in Tables 2 and 3; Fig. 1.

Pairwise comparisons confirmed significant early reductions (baseline to 1 month) in all cytokines for SRP, except IL-1β, which plateaued after the first month. In DLT-H, all cytokines declined significantly at both 1 and 3 months, with IL-1β dropping by more than 325 pg/mL by study end. While the incremental IL-1β reduction between months 1 and 3 did not reach significance, IL-6, IL-8, and TNF-α continued to decline. DLT-L showed significant decreases in IL-6, IL-8, and TNF-α at all intervals, though IL-1β reduction reached significance only at 3 months (Table 4).

Inter-group analysis demonstrated superior immunomodulation with DLT-H. At 3 months, DLT-H achieved the greatest reductions in pro-inflammatory cytokines, particularly IL-6 and TNF-α, whereas DLT-L produced intermediate improvements between DLT-H and SRP. Intergroup differences in the late interval (1–3 months) were less pronounced, although IL-6 reduction remained significantly greater in DLT-H versus SRP, and IL-8

Table 2 Effect of different treatment modalities on immune markers (IL-1β, IL-6, IL-8, and TNF-α) measured at baseline, 1 month, and 3 months after treatment for patients with periodontitis

	Mean	SD	Minimum	Maximum
Before treatment				
IL-1β	1721.11	293.77	1087.00	2427.00
IL-6	180.12	37.29	102.00	278.00
IL-8	276.36	101.11	116.00	492.00
TNF-α	216.78	39.21	117.00	290.00
After 1 month				
IL-1β	1596.63	322.76	994.00	2495.00
IL-6	157.10	35.38	89.00	241.00
IL-8	248.25	100.03	102.00	482.00
TNF-α	184.31	34.06	102.00	254.00
After 3 months				
IL-1β	1510.46	289.68	753.00	2335.00
IL-6	139.28	43.35	75.00	334.00
IL-8	224.14	95.71	93.00	453.00
TNF-α	162.80	33.58	95.00	235.00

IL-1β interleukin 1β, *IL-6* interleukin 6, *IL-8* interleukin 8, *TNF-α* Tumor necrosis factor-α

Table 3 Effect of different treatment modalities on immune markers (IL-1β, IL-6, IL-8, and TNF-α) measured at baseline (T0), 1 month (T1), and 3 months (T2) after treatment for patients with periodontitis

Parameter	Time of treatment			P-value
	T0	T1	T2	
SRP				
IL-1β	1722.93 ± 291.70	1613.04 ± 256.28	1558.74 ± 215.41	0.000
IL-6	172.22 ± 37.08	162.85 ± 36.68	150.07 ± 33.47	0.000
IL-8	272.41 ± 100.72	251.67 ± 97.39	236.04 ± 91.64	0.000
TNF	210.19 ± 38.19	189.52 ± 33.73	171.48 ± 33.93	0.000
DLT-H				
IL-1β	1725.56 ± 315.10	1499.19 ± 310.03	1400.52 ± 347.10	0.000
IL-6	184.48 ± 38.46	146.48 ± 27.08	119.44 ± 23.37	0.000
IL-8	261.07 ± 100.33	215.93 ± 89.22	190.56 ± 82.13	0.000
TNF	226.33 ± 39.24	182.07 ± 34.96	154.37 ± 30.99	0.000
DLT-L				
IL-1β	1714.85 ± 284.78	1677.67 ± 376.55	1572.11 ± 269.06	0.036
IL-6	183.67 ± 36.42	161.96 ± 39.97	148.33 ± 59.17	0.001
IL-8	295.59 ± 102.97	277.15 ± 106.66	245.81 ± 106.00	0.000
TNF	213.81 ± 39.79	181.33 ± 34.16	162.56 ± 34.71	0.000

IL-1β interleukin 1β, *IL-6* interleukin 6, *IL-8* nterleukin 8, *TNF-α* Tumor necrosis factor-α, *SRP* scaling and root planning, *DLT-L* Low-intensity diode laser, *DLT-H* high-intensity diode laser. Baseline (T0), 1 month (T1), and 3 months (T2) after treatment

reduction was significant in DLT-L versus SRP as shown in Tables 5, 6 and 7; Fig. 2.

Bland–Altman analyses confirmed the consistent superiority of DLT-H across all cytokines. In the early phase (baseline–1 month), DLT-H achieved significantly greater reductions than both SRP and DLT-L. Over the full 3-month period, DLT-H maintained significant superiority over SRP, whereas DLT-L effects were smaller and less consistent. Late-phase reductions were mainly significant for IL-6 and TNF-α in favor of DLT-H (Tables 8 and 9).

In summary, while all treatments reduced pro-inflammatory cytokines, high-intensity diode laser adjunctive therapy provided the most substantial and sustained immunological benefits compared with SRP alone or SRP with low-intensity laser.

Discussion

Chronic inflammatory periodontitis affects the tissues of the mouth and is characterized by dysbiotic biofilms and an exaggerated host immune response [2, 26]. While SRP effectively reduces bacterial biofilm and inflammation, its long-term efficacy can be limited, leading to interest in adjunctive therapies like diode laser treatment, which enhances healing through antimicrobial action and immune modulation [27, 28].

Despite accumulating evidence, the literature remains heterogeneous, with considerable variability in laser parameters and a paucity of studies directly comparing DLT-H and DLT-L under standardized conditions. Given the central role of cytokines such as IL-1β, IL-6, IL-8, and TNF-α in periodontal inflammation, the present study aimed to directly compare the adjunctive immunological effects of DLT-H and DLT-L with SRP in managing periodontitis.

This study rejected the null hypothesis, demonstrating that DLT-H as an adjunct to SRP produces significantly greater and more sustained reductions in pro-inflammatory cytokines than SRP alone or SRP with DLT-L. DLT-L demonstrated statistically significant improvements in periodontal parameters compared with baseline and provided additional benefits over SRP alone; however, the magnitude of improvement was less pronounced than that achieved with DLT-H, suggesting a more limited clinical benefit, which may reflect differences in the immunomodulatory effects associated with each laser modality.

This study demonstrates that nonsurgical periodontal therapy, whether performed as SRP alone or combined with diode laser adjuncts, effectively attenuates key pro-inflammatory mediators in GCF, confirming its central role in modulating the host immune response. Across all treatment modalities, significant reductions in IL-1β, IL-6, IL-8, and TNF-α were observed, reflecting suppression of destructive inflammatory pathways responsible for neutrophil recruitment, osteoclast activation, and matrix degradation. The marked decreases in IL-6 and TNF-α, in particular, underscore effective control over cytokines that drive alveolar bone resorption and connective tissue breakdown. Similarly, sustained declines in IL-1β and IL-8 indicate a reduced overall inflammatory

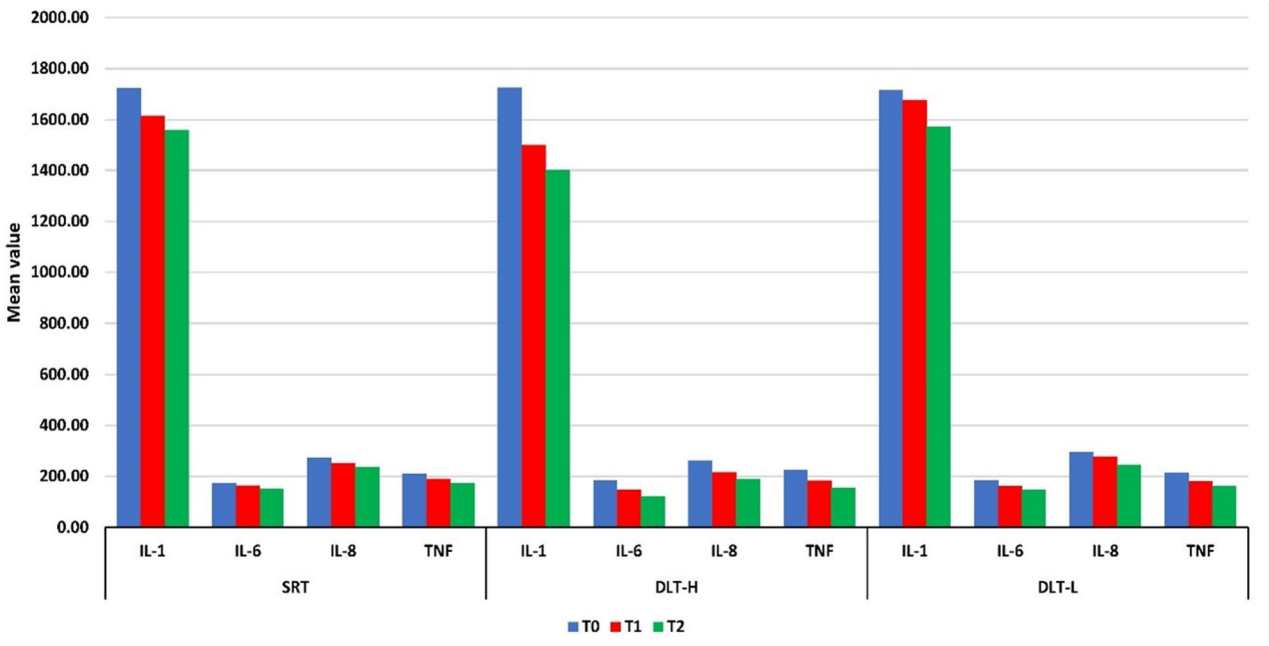


Fig. 1 Changes in gingival crevicular fluid levels of IL-1β (interleukin-1β), IL-6 (interleukin-6), IL-8 (interleukin-8), and TNF-α (tumor necrosis factor-α) in the SRP, DLT-H, and DLT-L groups at baseline (T0), 1 month (T1), and 3 months (T2)

Table 4 Pairwise comparisons between the treatment times for DLT-H, DLT-L and SRP treatment on immune markers (IL-1β, IL-6, IL-8, and TNF-α) measured at baseline, 1 month, and 3 months after treatment for patients with periodontitis

Treatment	Time		DLT-H		DLT-L		SRP	
			MD (95% CI)	P	MD (95% CI)	P	MD (95% CI)	P
IL-1β	T0	T1	226.37 (128.26, 324.48)	0	37.19 (-81.97, 156.34)	1	109.89 (46.33, 173.44)	0
		T2	325.04 (158.60, 491.47)	0	142.74 (2.66, 282.82)	0.045	164.19 (69.78, 258.59)	0
	T1	T2	98.67 (-42.34, 239.67)	0.255	105.56 (-51.78, 262.89)	0.294	54.30 (-6.62, 115.22)	0.093
IL-6	T0	T1	38 (17.79, 58.21)	0	21.70 (15.92, 27.49)	0	9.37 (6.17, 12.57)	0
		T2	65.04 (48.90, 81.17)	0	35.33 (12.35, 58.31)	0.002	22.15 (16.63, 27.67)	0
	T1	T2	27.04 (14.15, 39.92)	0	13.63 (-8.38, 35.64)	0.376	12.78 (8.25, 17.31)	0
IL-8	T0	T1	45.15 (26.38, 63.92)	0	18.44 (9.98, 26.91)	0	20.74 (7.12, 34.36)	0.002
		T2	70.52 (47.55, 93.48)	0	49.78 (35.05, 64.50)	0	36.37 (20.21, 52.54)	0
	T1	T2	25.37 (18.41, 32.33)	0	31.33 (20.66, 42.01)	0	15.63 (4.57, 26.69)	0.004
TNF	T0	T1	44.26 (36.06, 52.46)	0	32.48 (23.33, 41.63)	0	20.67 (12.56, 28.78)	0
		T2	71.96 (59.56, 84.37)	0	51.26 (39.40, 63.12)	0	3870 (29.24, 48.16)	0
	T1	T2	27.70 (19.62, 35.78)	0	18.78 (12.75, 24.81)	0	18.04 (13.66, 22.41)	0

burden and potential improvement in tissue homeostasis [18, 29].

Among the interventions, DLT-H produced the most rapid and durable immunological benefits. Cytokine levels, especially IL-6 and TNF-α, decreased more markedly and were maintained throughout the study period compared with both SRP alone and DLT-L. These effects likely result from DLT-H's combined photothermal and photobiomodulatory actions, which facilitate microbial reduction and direct modulation of host inflammatory pathways. The superior efficacy of DLT-H observed here aligns with previous reports that employed comparable specifications of diode laser irradiation demonstrating accelerated resolution of inflammation and enhanced periodontal healing when lasers are used adjunctively [16, 18, 22, 30–32]. In contrast, Lopes et al. reported no significant differences in IL-1β levels among the groups at follow-up, which may be attributed to differences in laser type, irradiation parameters, and treatment protocols compared with the present study [24, 33].

SRP alone yielded statistically significant but comparatively modest reductions in cytokine concentrations, with the greatest improvements noted at 1-month post-treatment [28]. While IL-6 and TNF-α continued to decline at 3 months, the limited further reduction in IL-1β suggests residual subclinical inflammation, consistent with prior findings highlighting the constraints of mechanical debridement in achieving sustained immunological

Table 5 Comparative analysis of immunological changes between the three treatment modalities

Parameter	Type of treatment			P-value
	SRP	DLT-H	DLT-L	
Δ (T0-T1)				
IL1β	109.89 ± 129.05	226.37 ± 199.22	37.19 ± 241.95	0.003
IL6	9.37 ± 6.51	38.00 ± 41.04	21.70 ± 11.75	0.000
IL8	20.74 ± 27.65	45.15 ± 38.12	18.44 ± 17.19	0.000
TNF	20.67 ± 16.47	44.26 ± 16.64	32.48 ± 18.58	0.000
Δ (T0-T2)				
IL1	164.19 ± 191.70	325.04 ± 337.96	142.74 ± 284.44	0.049
IL6	22.15 ± 11.21	65.04 ± 32.77	35.33 ± 46.67	0.000
IL8	36.37 ± 32.82	70.52 ± 46.63	49.78 ± 29.90	0.002
TNF	38.70 ± 19.21	71.96 ± 25.19	51.26 ± 24.08	0.000
Δ (T1-T2)				
IL1	54.30 ± 123.71	98.67 ± 286.32	105.56 ± 319.47	0.062
IL6	12.78 ± 9.20	27.04 ± 26.16	13.63 ± 44.70	0.005
IL8	15.63 ± 22.46	25.37 ± 14.13	31.33 ± 21.67	0.018
TNF	18.04 ± 8.88	27.70 ± 16.41	18.78 ± 12.25	0.045

IL-1β interleukin 1β, *IL-6* interleukin 6, *IL-8* interleukin 8, *TNF-α* Tumor necrosis factor-α, *SRP* scaling and root planning, *DLT-L* Low-intensity diode laser, *DLT-H* high-intensity diode laser. Baseline (T0), 1 month (T1), and 3 months (T2) after treatment

resolution [29]. These results emphasize the need for adjunctive interventions in cases with higher inflammatory burden.

DLT-L produced measurable immunological improvements, though these were generally delayed and less pronounced than those observed with DLT-H. Significant reductions were primarily observed at the 3-month follow-up, suggesting that low-intensity laser therapy exerts its effects through photobiomodulation, promoting mitochondrial activity, cellular repair, and gradual cytokine modulation rather than immediate microbial reduction [21, 34–36] While these findings support the use of DLT-L as an adjunctive therapy, its impact appears more suited to maintenance therapy or management of milder disease. DLT-L showed no significant cytokine differences compared to SRP ($p > 0.05$), indicating a slower immunological response, consistent with previous reports showing comparable GCF cytokine reductions between LLLT and SRP groups [34, 37, 38]. Conversely, some studies reported significantly lower cytokine levels in test sites compared to controls, reflecting variability in outcomes likely due to differences in laser peak power and fiber manipulation, which can alter tissue interactions and cytokine modulation [20, 35].

Pairwise analyses further confirmed the superior performance of DLT-H. During the early healing phase (baseline to 1 month), reductions in IL-6 and IL-8 were significantly greater with DLT-H than with SRP or DLT-L, indicating potent host-modulatory activity. In the later phase (1–3 months), DLT-H maintained its advantage by sustaining cytokine suppression, whereas DLT-L showed moderate, slower declines. These observations highlight the consistency and durability of the immunological effects induced by high-intensity laser treatment [18, 22, 37–39].

Collectively, these results suggest a clear hierarchy of efficacy: DLT-H provides the most robust and sustained immunomodulation, SRP alone is effective but less durable, and DLT-L offers slower, supportive benefits. The findings reinforce the biological rationale for using high-intensity lasers in cases with elevated inflammatory activity, while low-intensity lasers may serve as adjuncts in maintenance or milder disease scenarios.

Antimicrobial photodynamic therapy (aPDT) is another adjunctive approach that combines a

Table 6 Pairwise comparisons between the treatment times for different treatment methods on immune markers (IL-1β, IL-6, IL-8, and TNF-α) measured at baseline and 1 month after treatment for patients with periodontitis

Δ (T0-T1)						
Parameter	**Treatment**		**MD**	**95% CI for Difference**		**p-value**
				Lower	**Upper**	
IL-1β	SRP	DLT-H	−116.48	−246.79	13.83	0.015
		DLT-L	72.70	−57.61	203.01	1.000
	DLT-H	DLT-L	189.19	58.88	319.50	0.006
IL-6	SRP	DLT-H	−28.63	−45.23	−12.03	0.000
		DLT-L	−12.33	−28.94	4.27	0.001
	DLT-H	DLT-L	16.30	−0.31	32.90	0.075
IL-8	SRP	DLT-H	−24.41	−43.68	−5.13	0.000
		DLT-L	2.30	−16.98	21.57	1.000
	DLT-H	DLT-L	26.70	7.43	45.98	0.000
TNF	SRP	DLT-H	−23.59	−35.08	−12.10	0.000
		DLT-L	−11.82	−23.31	−0.32	0.032
	DLT-H	DLT-L	11.78	0.29	23.27	0.028

IL-1β interleukin 1β, *IL-6* interleukin 6, *IL-8* interleukin 8, *TNF-α* Tumor necrosis factor-α, *SRP* scaling and root planning, *DLT-L* low-intensity diode laser, *DLT-H* high-intensity diode laser. Baseline (T0) and 1 month (T1) after treatment

Table 7 Pairwise comparisons between the treatment times for different treatment methods on immune markers (IL-1β, IL-6, IL-8, and TNF-α) measured at baseline and 3 months after treatment for patients with periodontitis

Δ (T0-T2)						
Parameter	Treatment		MD	95% CI for Difference		P-value
				Lower	Upper	
IL-1β	SRP	DLT-H	− 160.85	− 345.98	24.28	0.088
		DLT-L	21.44	− 163.69	206.58	1.000
	DLT-H	DLT-L	182.30	− 2.83	367.43	0.118
IL-6	SRP	DLT-H	− 42.89	− 65.23	− 20.55	0.000
		DLT-L	− 13.19	− 35.53	9.16	0.002
	DLT-H	DLT-L	29.70	7.36	52.05	0.057
IL-8	SRP	DLT-H	− 34.15	− 58.90	− 9.39	0.001
		DLT-L	− 13.41	− 38.16	11.35	0.221
	DLT-H	DLT-L	20.74	− 4.02	45.50	0.234
TNF	SRP	DLT-H	− 33.26	− 48.56	− 17.96	0.000
		DLT-L	− 12.56	− 27.85	2.74	0.226
	DLT-H	DLT-L	20.70	5.41	36.00	0.008

IL-1β interleukin 1β, *IL-6* nterleukin 6, *IL-8* nterleukin 8, *TNF-α* Tumor necrosis factor-α, *SRP* scaling and root planning, *DLT-L* low-intensity diode laser, *DLT-H* high-intensity diode laser. baseline (T0) and 3 months (T2) after treatment

photosensitizer with light activation to reduce periodontal pathogens. Although aPDT has shown promising clinical and antimicrobial effects, diode laser therapy offers the advantage of not requiring exogenous photosensitizers while providing effective periodontal decontamination [40]. Beyond these local periodontal benefits, the observed reduction in pro-inflammatory cytokines may contribute to a decreased systemic inflammatory burden. Considering the recognized link between periodontitis and systemic inflammatory conditions, future studies incorporating systemic biomarkers are warranted to further explore these potential benefits.

Limitations of the study

This study has several limitations that should be acknowledged. The follow-up period of 3 months restricts the ability to evaluate long-term cytokine dynamics and the sustained effects of immune modulation. A longer observation period would provide a more comprehensive understanding of the temporal changes and potential stability of the observed responses. Moreover, the analysis was limited to a selected panel of pro-inflammatory cytokines, which may not fully represent the complexity of immune regulation involved in the healing process. Including a wider range of cytokines, growth factors, and anti-inflammatory mediators in future investigations would help clarify the broader biological mechanisms influenced by laser therapy. Future research should investigate emerging antimicrobial technologies, including alternating current and electrolytic decontamination systems, through rigorous laboratory and clinical studies to further improve periodontal infection control.

Additionally, the relatively small sample size may have limited the statistical power to detect minor but clinically relevant differences between the intervention groups. This could affect the generalizability of the findings to larger or more diverse populations. Future studies should consider recruiting larger cohorts, extending the follow-up duration, and employing more comprehensive immunological assessments to confirm these results and refine the understanding of laser-induced immunomodulation.

Conclusions

Within limitations of the present study, it could be concluded that all treatment modalities—SRP, DLT-H, and DLT-L—effectively reduced IL-1β, IL-6, IL-8, and TNF-α, demonstrating attenuation of periodontal inflammation. DLT-H produced the most pronounced and sustained cytokine suppression, reflecting its dual antimicrobial and immunomodulatory effects. SRP induced early but less durable reductions, while DLT-L provided gradual, meaningful immunological improvements consistent with photobiomodulation. These results support DLT-H as the superior adjunctive modality, with DLT-L offering potential benefits in maintenance or mild disease. Clinically, DLT-H may represent a valuable adjunctive option for managing periodontitis cases with a high inflammatory burden.

Future randomized clinical trials with larger sample sizes, extended follow-up, and comprehensive cytokine profiling are warranted to confirm the long-term clinical and immunological benefits of adjunctive diode laser therapy in periodontal treatment.

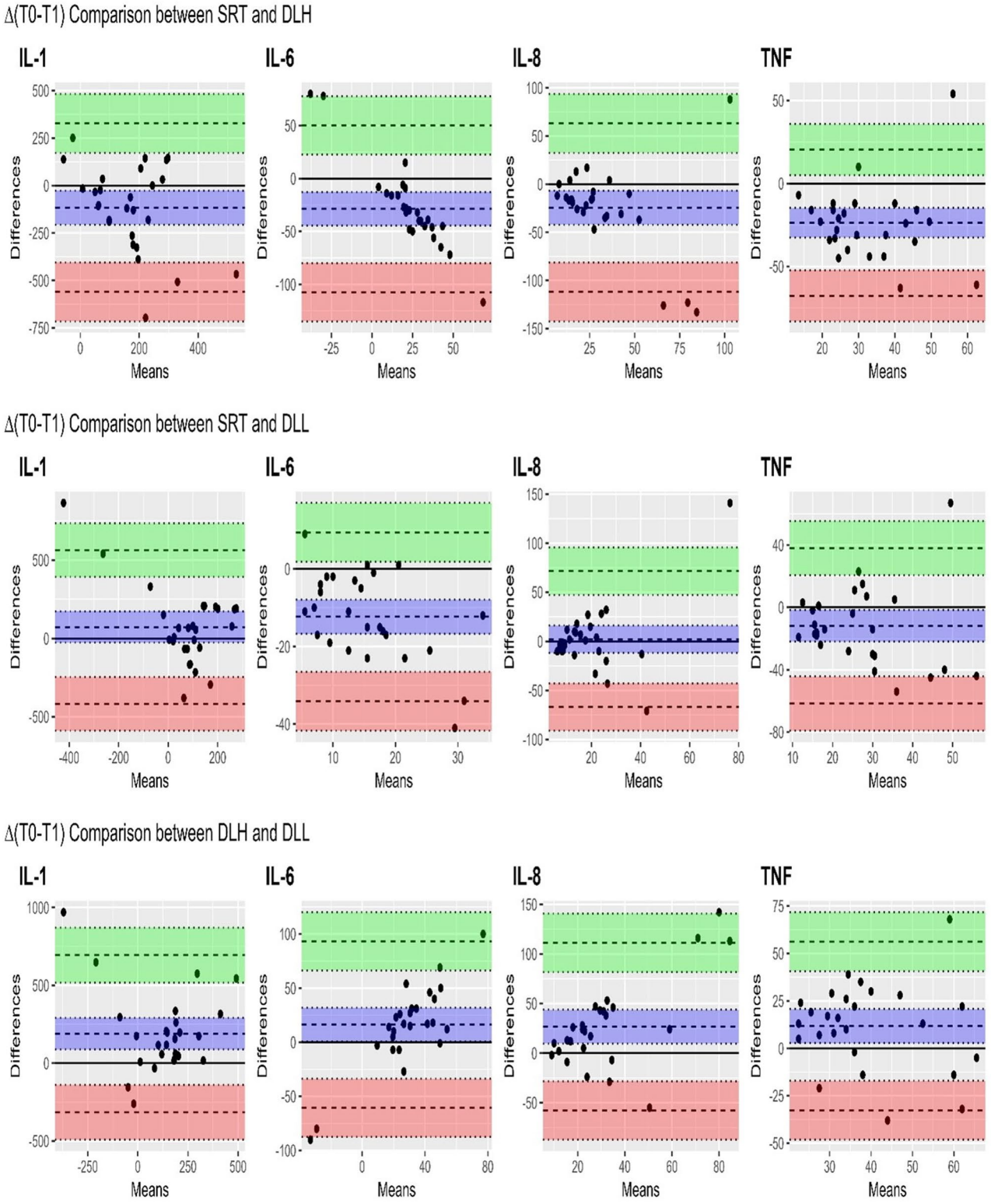


Fig. 2 Pairwise comparisons of changes in gingival crevicular fluid levels of IL-1β (interleukin-1β), IL-6 (interleukin-6), IL-8 (interleukin-8), and TNF-α (tumor necrosis factor-α) between baseline (T0) and 1 month (T1) following different treatment modalities in patients with periodontitis

Table 8 Pairwise comparisons of the effect of treatment times and different treatment modalities on immune markers (IL-1β, IL-6, IL-8, and TNF-α) measured 1 month and 3 months after treatment for patients with periodontitis

Δ (T1-T2)

Parameter	Treatment		MD	95% CI for Difference		P-value
				Lower	Upper	
IL-1β *	SRP	DLT-H	−44.37	−216.03	127.28	1.000
		DLT-L	−51.26	−222.91	120.40	1.000
	DLT-H	DLT-L	−6.89	−178.54	164.77	1.000
IL-6[á]	SRP	DLT-H	−14.26	−34.48	5.96	0.005
		DLT-L	−0.85	−21.08	19.37	0.056
	DLT-H	DLT-L	13.41	−6.82	33.63	1.000
IL-8[á]	SRP	DLT-H	−9.74	−22.91	3.43	0.131
		DLT-L	−15.70	−28.88	−2.53	0.018
	DLT-H	DLT-L	−5.96	−19.14	7.21	1.000
TNF[á]	SRP	DLT-H	−9.67	−18.25	−1.09	0.098
		DLT-L	−0.74	−9.32	7.84	1.000
	DLT-H	DLT-L	8.93	0.35	17.51	0.088

IL-1β interleukin 1β, *IL-6* interleukin 6, *IL-8* interleukin 8, *TNF-α* Tumor necrosis factor-α, *SRP* scaling and root planning, *DLT-L* low-intensity diode laser, *DLT-H* high-intensity diode laser. 1 month (T1), and 3 months (T2) after treatment

Table 9 Bland-Altman analysis for the comparison between treatment modalities

	IL-1		IL-6		IL-8		TNF	
	MD	*P*-value	MD	*P*-value	MD	*P*-value	MD	*P*-value
(T0-T1)								
SRP & DLH	−116.48	0.013	−28.63	0.001	−24.41	0.009	−23.59	< 0.001
SRP & DLL	72.70	0.144	−12.33	< 0.001	2.30	0.738	−11.81	0.023
DLH & DLL	189.19	0.001	16.30	0.040	26.70	0.003	11.78	0.012
(T0-T2)								
SRP & DLH	−160.85	0.036	−42.89	< 0.001	−34.15	0.001	−33.26	< 0.001
SRP & DLL	21.44	0.729	−13.19	0.175	−13.41	0.156	−12.56	0.059
DLH & DLL	182.30	0.058	29.70	0.002	20.74	0.069	20.70	0.006
(T1-T2)								
SRP & DLH	−44.37	0.433	−14.26	0.015	−11.41	0.077	−9.67	0.017
SRP & DLL	−51.26	0.446	−0.85	0.925	−15.70	0.002	−0.74	0.802
DLH & DLL	−6.89	0.941	13.41	0.216	−5.96	0.295	8.93	0.054

IL-1β interleukin 1β, *IL-6* interleukin 6, *IL-8* interleukin 8, *TNF-α* Tumor necrosis factor-α, *SRP* scaling and root planning, *DLT-L* low-intensity diode laser, *DLT-H* high-intensity diode laser (DLT-H). Baseline (T0), 1 month (T1), and 3 months (T2) after treatment

Acknowledgements
The authors would like to thank the managements of the Collage of Dentistry and hospital of University of Science and Technology for their assistance and supports.

Author contributions
AA, and WA contributed to the concept of the research, study design, data collection, supervision, statistical analysis, writing the original draft, and reading and editing the final paper. AAM contributed to supervision and writing the original draft. Every author evaluated and approved the final manuscript.

Funding
No funding.

Data availability
Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent participate
The research protocol received ethical approval from the University Research Ethics Committee (UREC) at the University of Science and Technology (UST), Sana'a, Yemen (Approval No. 1446/0016/UREC/UST). Written informed consent was obtained from all participants before participation in the study. All procedures were conducted in compliance with the principles outlined in the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 10 October 2025 / Accepted: 20 July 2026
Published online: 25 July 2026

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